Electrolyte : Lithium salt in an organic solvent

At the anode oxidation occurs

$Li\ (s) \rightarrow Li^{+}\ (aq) + e^{-}$

At the cathode reduction occurs

$Li^{+} + CoO_2\ (s) + e^{-} \rightarrow Li\ CoO_2\ (s)$

Overall reactions

$Li\ (s) + CoO_2 \rightarrow LiCoO_2\ (s)$

Both electrodes allow $Li^{+}$ ions to move in and out of their structures.

During discharge, the $Li^{+}$ ions produced at the anode move towards cathode through the non – aqueous electrolyte. When a potential greater than the emf produced by the cell is applied across the electrode, the cell reaction is reversed and now the $Li^{+}$ ions move from cathode to anode where they become embedded on the porous graphite electrode. This is known as intercalation.

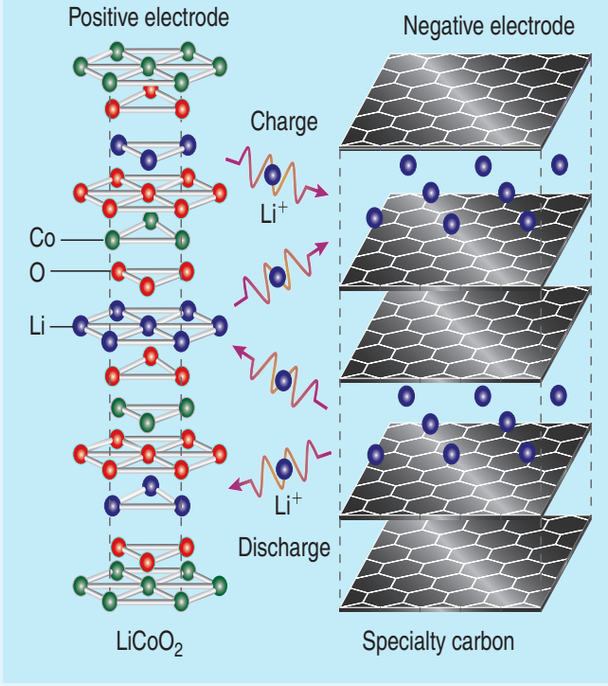



**Figure 9.13 Li-ion battery**

## Uses :

Used in cellular phones, laptops, computers, digital cameras, etc…

## Fuel cell

The galvanic cell in which the energy of combustion of fuels is directly converted into electrical energy is called the fuel cell. It requires a continuous supply of reactant to keep functioning. The general representation of a fuel cell is follows

Fuel | Electrode | Electrolyte | Electrode | Oxidant

Let us understand the function of fuel cell by

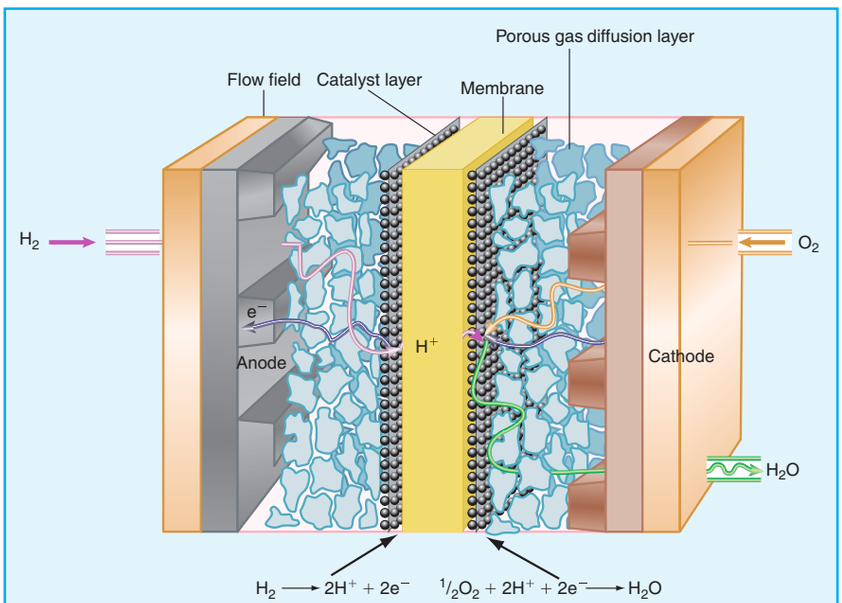



**Figure 9.14 $H_2O_2$ fuel cell**



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